

Amazon S3 Lifecycle Rules – Part 1 & 2

To Begin with the Lab

Summary of the Lab

In these two parts, **Amazon S3 Lifecycle Rules** were used to automate object storage management and reduce cost by moving objects through different storage classes over time. The first part explained how lifecycle rules can be applied to all objects or filtered using prefixes, object tags, or object size, so the rule can target only the right data. The second part added more detail by showing how lifecycle rules work with versioning, including current and non-current versions, storage class transitions after specific day counts, and expiration of objects after long retention periods. It also showed how delete markers can be cleaned up and how incomplete multipart uploads are handled separately. Together, the lab demonstrated how S3 can manage data automatically instead of relying on manual object changes.

Understanding Amazon S3 Lifecycle Rules

Amazon S3 Lifecycle Rules automate how objects are moved, archived, or deleted over time. They exist so you do not have to manually change the storage class of thousands of objects or clean up old versions one by one. In this lab, lifecycle rules were shown as the best way to move frequently used data from **Standard** to cheaper classes like **Standard-IA**, then to archive classes such as **Glacier Flexible Retrieval** and **Glacier Deep Archive**. When versioning is enabled, lifecycle rules can also act on current and non-current versions, which helps keep only the useful data while safely handling delete markers and old versions.

Example:

A company stores file1.txt in S3 and accesses it heavily for the first 30 days. After that, the lifecycle rule moves it to **Standard-IA**, then to **One Zone-IA** after 90 days, then to **Glacier Flexible Retrieval** after one year, and finally to **Glacier Deep Archive** after three years. If versioning is enabled, the rule can also expire the current version after 3650 days and delete the non-current version one day later.

Lab Steps

Step 1 – Review manual storage class changes

- Open the **Amazon S3** bucket.
- Upload an object and choose a storage class during upload by expanding **Properties** tab.

Properties

Specify storage class, encryption settings, tags, and more.

Storage class [Info](#)

Amazon S3 offers a range of storage classes designed for different use cases. [Learn more](#) or see [Amazon S3 pricing](#)

Storage class	Designed for	Bucket type	Availability Zones	Min storage duration	Min billable object size	Monitoring and auto-tiering fees	Retrieval fees
☒ Standard	Frequently accessed data (more than once a month) with milliseconds access	General purpose	≥ 3	-	-	-	-
☐ Intelligent-Tiering	Data with changing or unknown access patterns	General purpose	≥ 3	-	-	Per-object fees apply for objects ≥ 128 KB	-
☐ Standard-IA	Infrequently accessed data (once a month) with milliseconds access	General purpose	≥ 3	30 days	128 KB	-	Per-GB fees apply
☐ One Zone-IA	Recreatable, infrequently accessed data (once a month) with milliseconds access	General purpose or directory	1	30 days	128 KB	-	Per-GB fees apply
☐ Glacier Instant Retrieval	Long-lived archive data accessed once a quarter with instant retrieval in milliseconds	General purpose	≥ 3	90 days	128 KB	-	Per-GB fees apply
☐ Glacier Flexible Retrieval (formerly Glacier)	Long-lived archive data accessed once a year with retrieval of minutes to hours	General purpose	≥ 3	90 days	-	-	Per-GB fees apply

- Select an existing object and use **Object actions** to edit its storage class manually.

s3-version-disabled [Info](#)

Objects | Metadata | Properties | Permissions | Metrics | Management | File systems - new | Access Points

Objects (1/4)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant permissions.

Find objects by prefix

Name	Type	Last modified	Size
☒ file1.txt	txt	July 18, 2026, 18:52:03 (UTC+05:30)	
☐ windows_365_cloud_pc_microsoft_3-wallpaper-1920x1080.jpg	jpg	July 18, 2026, 18:52:59 (UTC+05:30)	2
☐ windows-11-blue-3840x2400-11776.png	png	July 18, 2026, 18:52:33 (UTC+05:30)	
☐ windows-11-waves-3840x2400-11767.png	png	July 18, 2026, 18:52:58 (UTC+05:30)	

Actions dropdown menu:

- Download as
- Share with a presigned URL
- Calculate total size
- Copy
- Move
- Initiate restore
- Query with S3 Select
- Edit actions
- Rename object
- Edit storage class**
- Edit server-side encryption
- Edit tags
- Make public using ACL

- Notice that manual changes are fine for one object but not practical for thousands.

Step 2 – Create a lifecycle rule

- Open the bucket **Management** tab.
- Click **Create lifecycle rule**.

The screenshot shows the AWS Management Console for a bucket named 's3-version-disabled'. The 'Management' tab is selected. Under 'Lifecycle configuration', there is a section for 'Lifecycle rules' and another for 'Replication rules (0)'. In the 'Lifecycle rules' section, a message states 'No lifecycle rules' and a 'Create lifecycle rule' button is highlighted with a red box. The 'Replication rules' section also has a 'Create replication rule' button.

- Enter a rule name such as Rule-1.
- Choose whether to apply the rule to all objects or only filtered objects.

The screenshot shows the 'Create lifecycle rule' page. The 'Lifecycle rule configuration' section is active. The 'Lifecycle rule name' field contains 'Rule-1'. Under 'Choose a rule scope', the 'Limit the scope of this rule using one or more filters' option is selected. The 'Filter type' is set to 'Prefix'. The 'Prefix' field contains an empty string. There are also sections for 'Object tags' and 'Object size'.

- Acknowledge the rule settings and continue.

Step 3 – Filter objects with prefix, tags, or size

- Use a **prefix** such as <FOLDER_NAME>/ to apply the rule only to objects in that folder path.
- Use an empty prefix or "" to target only root objects in the bucket.

Create lifecycle rule [Info](#)

Lifecycle rule configuration

Lifecycle rule name
Rule-1
Up to 255 characters

Choose a rule scope
☒ Limit the scope of this rule using one or more filters
☐ Apply to all objects in the bucket

Filter type
You can filter objects by prefix, object tags, object size, or whatever combination suits your usecase.

Prefix
Add filter to limit the scope of this rule to a single prefix.
/

Don't include the bucket name in the prefix. Using certain characters in key names can cause problems with some applications and protocols. [Learn more](#)

Object tags
You can limit the scope of this rule to the key/value pairs added below.
[Add tag](#)

Object size
You can limit the scope of this rule to apply to objects based on their size. [Learn more](#)
☐ Specify minimum object size
☐ Specify maximum object size

- Use **object tags** such as logs when you want the rule to match tagged objects and the tags can be set when uploading an object.

Specify an encryption key
The specified encryption key is used to encrypt objects before storing them in Amazon S3.

Checksums
Checksums are used for data integrity verification of new objects. [Learn more](#)

Checksum function
Checksum functions are used to calculate the checksum value. For objects smaller than 16 MB, only the full object checksum type is supported, for all checksum algorithms.
CRC64NVME (recommended)

Precalculated value - optional
When you provide a precalculated value for a single object, S3 compares it to the value it calculates using the selected checksum algorithm. If the values don't match, the upload will fail. [Learn more](#)
Enter value

Tags - optional
You can use object tags to analyze, manage, and specify permissions for objects. [Learn more](#)

Key	Value - optional
Key	Value

[Add tag](#) [Remove](#)

- Use **object size** when you want the rule to apply only to objects within a size range.

Amazon S3 > Buckets > s3-version-enabled > Lifecycle configuration > Create lifecycle rule

You can filter objects by prefix, object tags, object size, or whatever combination suits your usecase.

Prefix
Add filter to limit the scope of this rule to a single prefix.

Don't include the bucket name in the prefix. Using certain characters in key names can cause problems with some applications and protocols. [Learn more](#)

Object tags
You can limit the scope of this rule to the key/value pairs added below.

Object size
You can limit the scope of this rule to apply to objects based on their size. [Learn more](#)

☒ **Specify minimum object size**
 The filter will allow objects larger than the entered value.

 Must be a positive number 0 B or larger.

☒ **Specify maximum object size**
 The filter will allow objects smaller than the entered value.

 Must be a positive number 1.0 B or larger.

Unit selection:
 KB (selected)
 Bytes
 MB
 GB

Lifecycle rule actions
 Choose the actions you want this rule to perform.

Step 4 – Configure current-version storage transitions

- Apply the rule to the current version of objects.

Amazon S3 > Buckets > s3-version-enabled > Lifecycle configuration > Create lifecycle rule

Lifecycle rule actions
 Choose the actions you want this rule to perform.

☒ **Transition current versions of objects between storage classes**
 This action will move current versions.

☐ Transition noncurrent versions of objects between storage classes
 This action will move noncurrent versions.

☐ Expire current versions of objects

☐ Permanently delete noncurrent versions of objects

☐ Delete expired object delete markers or incomplete multipart uploads
 These actions are not supported when filtering by object tags or object size.

Transitions are charged per request
 For a lifecycle transition action, each request corresponds to an object transition. For details on lifecycle transition pricing, see requests pricing info on the [Storage & requests](#) tab of the [Amazon S3 pricing page](#).

☒ I acknowledge that this lifecycle rule will incur a transition cost per request.

By default, objects less than 128KB will not transition across any storage class
 We don't recommend transitioning objects less than 128 KB because the transition costs can outweigh the storage savings. If your use case requires transitioning objects less than 128 KB, specify a minimum object size filter for each applicable lifecycle rule with a transition action.

Transition current versions of objects between storage classes
 Choose transitions to move current versions of objects between storage classes based on your use case scenario and performance access requirements. These transitions start from when the objects are created and are consecutively applied. [Learn more](#)

Choose storage class transitions
 Standard-IA (selected)

Days after object creation

- Move objects from **Standard** to **Standard-IA** after 30 days.

applicable lifecycle rule with a transition action.

Transition current versions of objects between storage classes

Choose transitions to move current versions of objects between storage classes based on your use case scenario and performance access requirements. These transitions start from when the objects are created and are consecutively applied. [Learn more](#)

Choose storage class transitions

Standard-IA

Days after object creation

30

Remove

Add transition

Review transition and expiration actions

Current version actions

Day 0

- Objects uploaded

↓

Day 30

- Objects move to Standard-IA

Noncurrent versions actions

Day 0

No actions defined.

Cancel Create rule

- Move objects from **Standard-IA** to **One Zone-IA** after 90 days.

applicable lifecycle rule with a transition action.

Transition current versions of objects between storage classes

Choose transitions to move current versions of objects between storage classes based on your use case scenario and performance access requirements. These transitions start from when the objects are created and are consecutively applied. [Learn more](#)

Choose storage class transitions

Standard-IA

Days after object creation

30

Remove

One Zone-IA

90

Remove

Add transition

Review transition and expiration actions

Current version actions

Day 0

- Objects uploaded

↓

Day 30

- Objects move to Standard-IA

↓

Day 90

- Objects move to One Zone-IA

Noncurrent versions actions

Day 0

No actions defined.

Cancel Create rule

- Move objects to **Glacier Flexible Retrieval** after one year.

Transition current versions of objects between storage classes

Choose transitions to move current versions of objects between storage classes based on your use case scenario and performance access requirements. These transitions start from when the objects are created and are consecutively applied. [Learn more](#)

Choose storage class transitions	Days after object creation	
Standard-IA	30	Remove
One Zone-IA	90	Remove
Glacier Flexible Retrieval (formerly Glacier)	365	Remove

[Add transition](#)

Review transition and expiration actions

Current version actions

Day 0

- Objects uploaded

↓

Day 30

- Objects move to Standard-IA

↓

Noncurrent versions actions

Day 0

No actions defined.

- Move objects to **Glacier Deep Archive** after three years (1095 days).

Transition current versions of objects between storage classes

Choose transitions to move current versions of objects between storage classes based on your use case scenario and performance access requirements. These transitions start from when the objects are created and are consecutively applied. [Learn more](#)

Choose storage class transitions	Days after object creation	
Standard-IA	30	Remove
One Zone-IA	90	Remove
Glacier Flexible Retrieval (formerly Glacier)	365	Remove
Glacier Deep Archive	1095	Remove

[Add transition](#)

Review transition and expiration actions

Current version actions

Day 0

- Objects uploaded

↓

Day 30

- Objects move to Standard-IA

↓

Day 90

- Objects move to One Zone-IA

↓

Noncurrent versions actions

Day 0

No actions defined.

Step 5 – Configure expiration for current and non-current versions

- Edit the lifecycle rule again.
- Enable **Expire current version of object**.

Amazon S3 > Buckets > s3-version-disabled > Lifecycle configuration > Create lifecycle rule

Lifecycle rule actions

Choose the actions you want this rule to perform.

- ☒ **Transition current versions of objects between storage classes**
This action will move current versions.
- ☐ Transition noncurrent versions of objects between storage classes
This action will move noncurrent versions.
- ☒ **Expire current versions of objects**
- ☐ Permanently delete noncurrent versions of objects
- ☐ Delete expired object delete markers or incomplete multipart uploads
These actions are not supported when filtering by object tags or object size.

Transitions are charged per request
For a lifecycle transition action, each request corresponds to an object transition. For details on lifecycle transition pricing, see requests pricing info on the [Storage & requests](#) tab of the [Amazon S3 pricing page](#).

☒ I acknowledge that this lifecycle rule will incur a transition cost per request.

By default, objects less than 128KB will not transition across any storage class
We don't recommend transitioning objects less than 128 KB because the transition costs can outweigh the storage savings. If your use case requires transitioning objects less than 128 KB, specify a minimum object size filter for each applicable lifecycle rule with a transition action.

Transition current versions of objects between storage classes

Choose transitions to move current versions of objects between storage classes based on your use case scenario and performance access requirements. These transitions start from when the objects are created and are consecutively applied. [Learn more](#)

Choose storage class transitions: Standard-IA
Days after object creation: 30
[Remove](#)

- Set expiration for 3650 days.

Amazon S3 > Buckets > s3-version-disabled > Lifecycle configuration > Create lifecycle rule

Expire current versions of objects

For version-enabled buckets, Amazon S3 adds a delete marker and the current version of an object is retained as a noncurrent version. For non-versioned buckets, Amazon S3 permanently removes the object. [Learn more](#)

Days after object creation: 3650

Review transition and expiration actions

Current version actions	Noncurrent versions actions
Day 0 Objects uploaded	Day 0 No actions defined.
↓	
Day 30 Objects move to Standard-IA	
↓	
Day 90 Objects move to One Zone-IA	
↓	
Day 365	

- Enable Permanently delete non-current version of object.

Amazon S3 > Buckets > s3-version-disabled > Lifecycle configuration > Create lifecycle rule

Lifecycle rule actions

Choose the actions you want this rule to perform.

- ☒ Transition current versions of objects between storage classes
This action will move current versions.
- ☐ Transition noncurrent versions of objects between storage classes
This action will move noncurrent versions.
- ☒ Expire current versions of objects
- ☒ Permanently delete noncurrent versions of objects
- ☐ Delete expired object delete markers or incomplete multipart uploads
These actions are not supported when filtering by object tags or object size.

Transitions are charged per request
For a lifecycle transition action, each request corresponds to an object transition. For details on lifecycle transition pricing, see requests pricing info on the [Storage & requests](#) tab of the [Amazon S3 pricing page](#).

☒ I acknowledge that this lifecycle rule will incur a transition cost per request.

By default, objects less than 128KB will not transition across any storage class
We don't recommend transitioning objects less than 128 KB because the transition costs can outweigh the storage savings. If your use case requires transitioning objects less than 128 KB, specify a minimum object size filter for each applicable lifecycle rule with a transition action.

Transition current versions of objects between storage classes

Choose transitions to move current versions of objects between storage classes based on your use case scenario and performance access requirements. These transitions start from when the objects are created and are consecutively applied. [Learn more](#)

Choose storage class transitions: Standard-IA
Days after object creation: 30
[Remove](#)

- Set the non-current version deletion to one day after the object becomes non-current.

Amazon S3 > Buckets > s3-version-disabled > Lifecycle configuration > Create lifecycle rule

Glacier Deep Archive 1095 [Remove](#)

[Add transition](#)

Expire current versions of objects

For version-enabled buckets, Amazon S3 adds a delete marker and the current version of an object is retained as a noncurrent version. For non-versioned buckets, Amazon S3 permanently removes the object. [Learn more](#)

Days after object creation: 3650

Permanently delete noncurrent versions of objects

Choose when Amazon S3 permanently deletes specified noncurrent versions of objects. [Learn more](#)

Days after objects become noncurrent: 1

Number of newer versions to retain - Optional: Number of versions
Can be 1 to 100 versions. All other noncurrent versions will be moved.

Review transition and expiration actions

Current version actions	Noncurrent versions actions
Day 0 • Objects uploaded	Day 0 • Objects become noncurrent

Step 6 – Handle delete markers

- Review the **expired object delete marker** option.

Lifecycle rule actions
Choose the actions you want this rule to perform.

- ☒ Transition current versions of objects between storage classes
This action will move current versions.
- ☐ Transition noncurrent versions of objects between storage classes
This action will move noncurrent versions.
- ☒ Expire current versions of objects
- ☒ Permanently delete noncurrent versions of objects
- ☒ Delete expired object delete markers or incomplete multipart uploads.
These actions are not supported when filtering by object tags or object size.

Transitions are charged per request
For a lifecycle transition action, each request corresponds to an object transition. For details on lifecycle transition pricing, see requests pricing info on the [Storage & requests tab of the Amazon S3 pricing page](#).

☒ I acknowledge that this lifecycle rule will incur a transition cost per request.

By default, objects less than 128KB will not transition across any storage class
We don't recommend transitioning objects less than 128 KB because the transition costs can outweigh the storage savings. If your use case requires transitioning objects less than 128 KB, specify a minimum object size filter for each applicable lifecycle rule with a transition action.

Transition current versions of objects between storage classes
Choose transitions to move current versions of objects between storage classes based on your use case scenario and performance access requirements. These transitions start from when the objects are created and are consecutively applied. [Learn more](#)

Choose storage class transitions: Standard-IA
Days after object creation: 30
[Remove](#)

- Understand that deleting an object in a versioned bucket creates a delete marker.
- If needed, remove the delete marker to restore the file.
- Let lifecycle cleanup remove leftover markers when the rule is configured for it.

Step 7 – Review incomplete multipart uploads

- Open the lifecycle rule settings again.
- Review the **incomplete multipart upload** option.

Permanently delete noncurrent versions of objects
Choose when Amazon S3 permanently deletes specified noncurrent versions of objects. [Learn more](#)

Days after objects become noncurrent: 1
Number of newer versions to retain - Optional: Number of versions (Can be 1 to 100 versions. All other noncurrent versions will be moved.)

Delete expired object delete markers or incomplete multipart uploads
Expired object delete markers
This action will remove expired object delete markers and may improve performance. An expired object delete marker is removed if all noncurrent versions of an object expire after deleting a versioned object. This action is not available when "Expire current versions of objects" is selected. [Learn more](#)

☒ Delete expired object delete markers
You cannot enable Delete expired object delete markers if you enable Expire current versions of objects.

Incomplete multipart uploads
This action will stop all incomplete multipart uploads, and the parts associated with the multipart upload will be deleted. [Learn more](#)

☒ Delete incomplete multipart uploads
Number of days: 1
Integer must be greater than 0.

Review transition and expiration actions
Current version actions:
Noncurrent versions actions:
Days after object creation:
Days after object creation:

- Keep this for later multipart upload handling, since it is a separate S3 concept.

Verification

- Verified that lifecycle rules automate S3 object management.
- Verified that lifecycle rules can filter objects using prefix, tags, and object size.

- Verified that current-version objects can transition through storage classes automatically.
- Verified that versioned buckets can expire current versions after a fixed number of days.
- Verified that non-current versions can be deleted separately.
- Verified that delete markers are part of versioned object deletion behavior.
- Verified that lifecycle rules can support long-term cost optimization for S3 data.